

Project Catalyst

Product & Architecture Overview

What it is: a self-contained React sandbox for paid-media creative testing. A marketer describes an offer, writes competing ad hooks, assigns each hook one of seven synthetic buyer personas, and gets an instant scored critique of every hook through its persona's lens — before a dollar hits a live auction. It runs entirely client-side, with an optional local backend for live Claude scoring.

1. Executive Summary

Creative testing in paid media is slow and blunt: teams wait days for live data, and judge copy against a single fuzzy “audience.” Catalyst makes a sharper workflow tangible — it shows how a marketer could pressure-test a set of ad hooks against distinct buyer mindsets and get a per-hook, per-persona read in seconds.

Important framing: by default the persona scoring is a deterministic rule engine — keyword-and-pattern heuristics, not a machine-learning model — so the hosted demo is free, instant and perfectly reproducible. The scoring engine is swappable behind a clean interface, and the project ships a working Claude-backed evaluator that runs the same UI on real model calls when enabled. The value sits in the system design, stated plainly, not in a claim the code can't back up.

2. The Problem and the Demonstration

What's actually broken	What Catalyst demonstrates
Slow feedback loops. Teams wait days for live performance data before they can tell a weak ad hook from a strong one.	Pre-flight critique. Each hook is scored instantly through an assigned buyer-persona lens, so weak angles get cut before any spend.
One-size persona testing. Creative is judged against a single vague “audience,” missing how different buyer mindsets react.	Targeted persona mapping. Pick which of seven personas are in play, then assign the right one to each hook for a sharper read.

3. Functional Modules

I. Creative Input Board

The intake panel captures what a marketer would provide and how they want it tested.

- **Brief:** product/service and a short audience description.
- **Dynamic hooks:** add or remove ad angles (1–6), each with its own assigned persona.
- **Active persona set:** choose which of the seven personas are in play, then map one to each hook.

- **Sample loader:** one click loads a ready-made ad concept — brief, hooks, persona mapping and a creative image.

II. Synthetic Persona Evaluation

Each hook is scored 0–100 through the single persona assigned to it, with a one-line rationale. The seven lenses:

- **The Skeptic (CFO):** cost, ROI and quantifiable proof; penalizes vague hype.
- **The Impulse Buyer (executive):** speed-to-value, bold hooks, emotional pull.
- **The Feature Critic (product):** technical specifics, integrations, structural clarity.
- **The Bargain Hunter:** discounts, savings and obvious value; penalizes premium framing.
- **The Brand Loyalist:** reputation, reviews and social proof.
- **The Researcher:** data, comparisons and detailed, cited evidence.
- **The Trend Seeker:** novelty, FOMO and momentum; penalizes legacy framing.

III. Sample Ads

Three brand-styled presets make the tool demo-ready out of the box, each loading a brief, persona mapping, hooks and a Canva-generated creative image:

- **Ledgerly** — expense automation (B2B SaaS).
- **Drift** — a weighted cooling blanket (DTC consumer).
- **Capital Pulse** — instant working-capital lines (fintech).

4. Architecture

- **Stack:** Vite + React 19 + TypeScript + Tailwind CSS v4, styled to the Mansa Tech brand. Client-side, with an optional local Express backend for the Claude path.
- **Swappable engine:** the rule engine and the Claude evaluator both implement one `PersonaEvaluator` interface; switching backends is a single line, and the UI never changes.
- **Deterministic by default:** the keyless static deploy runs the rule engine — zero latency, zero cost, identical results every run.
- **Quality gates:** Vitest unit tests cover the scoring rules and the server's score normalization; GitHub Actions runs install, lint, test and build on every push.
- **Deploy:** version-controlled in Git/GitHub and auto-deployed to Render on push. The live site runs the rule engine; the Claude backend is a local, key-held option.

5. Known Limitations and Next Steps

Stated up front, because naming the edges of a system reads as competence, not weakness.

- **Default scoring is heuristic.** A Claude-backed evaluator exists but runs locally. Next: host the backend so the live demo can score with the model.
- **Rule keyword lists are tunable placeholders.** Next: refine them, or lean on the LLM path for nuance.
- **No persistence.** Next: add a thin API + datastore so briefs, hooks and results can be saved and compared.

- **Sample images are static stills.** Next: generate per-hook creative variants to evaluate alongside the copy.

6. Portfolio Positioning

Catalyst is a systems-design portfolio piece that is actually shipped: tested, gated by CI, deployed, and on-brand. It shows the ability to scope a real marketing-ops problem, model a credible solution end to end, and build it with the architecture seams (a swappable scoring backend) that let it grow — while being honest about which parts are real and which are simulated. The interface it presents and the seams it leaves open are the proof of skill; the deterministic internals are a conscious tradeoff to keep the demo free and reliable.